Thomas Blankers

Assistant Professor of Evolutionary Ecology

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POSITIONS

2022–present	**Assistant Professor of Evolutionary Ecology**, Institute for Biodiversity and Ecosystem Dynamics, University of Amsterdam
2021–2022	**Research Fellow, Origins Center**, University of Amsterdam. With Prof. Astrid T. Groot
2018–2020	**Marie Curie Postdoctoral Fellow**, University of Amsterdam. With Prof. Astrid T. Groot
2016–2018	**Postdoctoral Associate**, Department of Neurobiology and Behavior, Cornell University, Ithaca, NY, USA. With Prof. Kerry L. Shaw
2012–2016	**Guest Researcher**, Museum für Naturkunde Berlin, Leibniz Institute for Evolution and Biodiversity Science. GENART project (functional genomics of speciation)

EDUCATION

2016	**PhD**, Humboldt-Universität zu Berlin, *summa cum laude*. Thesis: *Acoustic communication, sexual selection, and speciation in field crickets.* Supervisor: Prof. R. Matthias Hennig
2011	**MSc Biological Sciences** (Ecology and Evolution), University of Amsterdam, *cum laude* Research project (60 EC), Stony Brook University, NY: *Adaptive radiation and morphological diversification in salamanders and lizards.* Supervisors: Prof. John J. Wiens, Prof. Steph B. Menken Research project (30 EC), VU Amsterdam: *Nutrient signaling: genomics underlying lipid metabolism pathways in the parasitoid wasp Nasonia vitripennis.* Supervisors: Dr. Merijn Kant, Prof. Jacintha Ellers, Dr. Bertanne Visser
2008	**BSc Biology**, University of Amsterdam Research project (18 EC): *Adaptive speciation appraised: how disruptive selection forces sympatric populations of the two-spotted spider mite to diversify.* Supervisor: Dr. Martijn Egas

FUNDING

2024	**NWA-ORC PRELIFE**: Pathways, Reactions, and Environments leading to LIFE (NWA.1630.23.013), **€6,750,000**. Co-applicant, partner on two subprojects, subquestion leader during consortium forming. Main applicant: Inge Loes ten Kate
2024	**Starters Grant**, Dutch Ministry of Education, Culture and Science, **€300,000**. Personal grant
2023	**EU COST Action CA22102 E-NICHE**: European Network in Chemical Ecology, **€600,000**. Co-applicant, working group leader. Main applicant: Anne-Geneviève Bagnères
2022	**NWO NWA small project** 1418.22.013, **€300,000**. Subproject leader; €100k to our project, co-applied with Astrid T. Groot and Sander G. van Doorn. Main applicant: Pedro Russo
2020	**NWO NWA small project** 1418.20.026, **€150,000**. Subproject leader; €50k to our project, co-applied with Karen Bisschop and the PIs of the Origins Center "predicting evolution" game changer. Main applicant: Inge Loes ten Kate
2018	**Marie Skłodowska-Curie Individual Fellowship**, **€181,000**
2011	Stichting Dr. Hendrik Muller's Vaderlandsch Fonds, **€700** · Amsterdams Universiteitsfonds, **€500**

PUBLICATIONS

‡ shared first authorship · * BSc/MSc student contribution · Full list with links at thomasblankers.com

27. Do the benefits of hybridization outweigh the costs under conditions of global change?
Krapf, P., **Blankers, T.** & Kulmuni, J. 2026. Preprint, EcoEvoRxiv. doi:10.32942/X2795S

26. Additive effects of environmental and demographic variation shape the repeatability of evolution across replicated experiments.
Bisschop, K., Wortel, M.T., Kraaijeveld, K., Mariën, J., Rathmann, I., … Groot, A.T., Visser, M.E. & **Blankers, T.** 2026. *Evolution Letters* 10: 382–395. doi:10.1093/evlett/qrag017

25. The impact of environmental pollution on dragonflies and damselflies (Odonata) and the role of climate change as an interactive stressor.
Guerra, D., **Blankers, T.**, Kraak, M.H.S. & Burdfield-Steel, E.R. 2026. *Environmental Toxicology and Chemistry* 45: 2056–2090. doi:10.1093/etojnl/vgag069

24. Resource limitation reveals that high acetate levels in *Heliothis subflexa* sex pheromone blend are associated with reduced fitness.
Fruitet, E., De Jong, R., **Blankers, T.**, Groot, A.T. & Burdfield-Steel, E.R. 2026. *Journal of Evolutionary Biology* 39: 137–147. doi:10.1093/jeb/voaf126

23. European chemical ecologists translate the language of life into sustainability.
Verheggen, F., Allmann, S., Stopka, P., Pappas, M., Avila, C., Prado, S., Romano, G., **Blankers, T.**, Vanderplanck, M., Colazza, S. & Bagnères, A.G. 2024. *Entomologia Generalis* 44: 477–478. doi:10.1127/entomologia/2024/2729

22. The biogeographic and evolutionary processes shaping population divergence in *Laupala*.
Blankers, T. & Shaw, K.L. 2024. *Molecular Ecology* 33: e17444. doi:10.1111/mec.17444

21. The evolutionary importance of intraspecific variation in sexual communication across sensory modalities.
Groot, A.T., **Blankers, T.**, Halfwerk, W. & Burdfield-Steel, E. 2024. *Annual Review of Entomology* 69: 21–40. doi:10.1146/annurev-ento-030223-111608

20. Quantitative trait loci underlying a speciation phenotype.
Waller, W.H.*, **Blankers, T.**, Xu, M. & Shaw, K.L. 2023. *Insect Molecular Biology* 32: 592–602. doi:10.1111/imb.12858

19. Towards evolutionary predictions: current promises and challenges.
Wortel, M.T., Agashe, D., Bailey, S.F., Bank, C., Bisschop, K., **Blankers, T.**, … Pennings, P.S. 2023. *Evolutionary Applications* 16: 3–21. doi:10.1111/eva.13513

18. Population bottleneck has only marginal effect on fitness evolution and its repeatability in dioecious *Caenorhabditis elegans*.
Bisschop, K.‡, **Blankers, T.‡**, Mariën, J., Wortel, M.T., Egas, M., Groot, A.T., Visser, M.E. & Ellers, J. 2022. *Evolution* 76: 1896–1904. doi:10.1111/evo.14556

17. Experimental evolution of a pheromone signal.
Blankers, T., Fruitet, E., Burdfield-Steel, E. & Groot, A.T. 2022. *Ecology and Evolution* 12: e8941. doi:10.1002/ece3.8941

16. Sex pheromone signal and stability covary with fitness.
Blankers, T., Lievers, R., Plata, C.*, van Wijk, M., van Veldhuizen, D. & Groot, A.T. 2021. *Royal Society Open Science* 8: 210180. doi:10.1098/rsos.210180

15. Parallel genomic architecture underlies repeated sexual signal divergence in Hawaiian *Laupala* crickets.
Blankers, T., Oh, K.P. & Shaw, K.L. 2019. *Proceedings of the Royal Society B* 286: 20191479. doi:10.1098/rspb.2019.1479

14. Physical linkage and mate preference generate linkage disequilibrium for behavioral isolation in two parapatric crickets.
Blankers, T.‡, Berdan, E.L.‡, Hennig, R.M. & Mayer, F. 2019. *Evolution* 73: 777–791. doi:10.1111/evo.13706

13. The genomic architecture of a rapid island radiation: recombination rate variation, chromosome structure, and genome assembly of the Hawaiian cricket *Laupala*.
Blankers, T., Oh, K.P., Bombarely, A. & Shaw, K.L. 2018. *Genetics* 209: 1329–1344. doi:10.1534/genetics.118.300894

12. The genetics of a behavioral speciation phenotype in an island system.
Blankers, T., Oh, K.P. & Shaw, K.L. 2018. *Genes* 9: 346. doi:10.3390/genes9070346

11. Codivergence but limited covariance of wing shape and calling song structure in field crickets (*Gryllus*).
Blankers, T., Block, R.* & Hennig, R.M. 2018. *Evolutionary Biology* 45: 144–155. doi:10.1007/s11692-017-9439-2

10. Demography and selection shape transcriptomic divergence in field crickets.
Blankers, T., Vilaça, S.T., Waurick, I., Gray, D.A., Hennig, R.M., Mazzoni, C.J., Mayer, F. & Berdan, E.L. 2018. *Evolution* 72: 553–567. doi:10.1111/evo.13435

9. Multivariate phenotypic evolution: divergent acoustic signals and sexual selection in *Gryllus* field crickets.
Blankers, T., Gray, D.A. & Hennig, R.M. 2017. *Evolutionary Biology* 44: 43–55. doi:10.1007/s11692-016-9388-1

8. Multivariate female preference tests reveal latent perceptual biases.
Gray, D.A., **Blankers, T.** & Hennig, R.M. 2016. *Proceedings of the Royal Society B* 283: 20161972. doi:10.1098/rspb.2016.1972

7. Acoustic communication, sexual selection, and speciation in field crickets.
Blankers, T. 2016. PhD thesis, Humboldt-Universität zu Berlin. edoc.hu-berlin.de/handle/18452/18205

6. A genes eye view of ontogeny: de novo assembly and profiling of the *Gryllus rubens* transcriptome.
Berdan, E.L., **Blankers, T.**, Waurick, I., Mazzoni, C.J. & Mayer, F. 2016. *Molecular Ecology Resources* 16: 1478–1490. doi:10.1111/1755-0998.12530

5. Divergence in male cricket song and female preference functions in three allopatric sister species.
Hennig, R.M., **Blankers, T.** & Gray, D.A. 2016. *Journal of Comparative Physiology A* 202: 347–360. doi:10.1007/s00359-016-1083-2

4. Phenotypic variation and covariation indicate high evolvability of acoustic communication in crickets.
Blankers, T., Lübke, A.K.* & Hennig, R.M. 2015. *Journal of Evolutionary Biology* 28: 1656–1669. doi:10.1111/jeb.12686

3. Conservation of multivariate female preference functions and preference mechanisms in three species of trilling field crickets.
Blankers, T., Hennig, R.M. & Gray, D.A. 2015. *Journal of Evolutionary Biology* 28: 630–641. doi:10.1111/jeb.12599

2. Contrasting global-scale evolutionary radiations: phylogeny, diversification, and morphological evolution in the major clades of iguanian lizards.
Blankers, T., Townsend, T.M., Pepe, K., Reeder, T.W. & Wiens, J.J. 2013. *Biological Journal of the Linnean Society* 108: 127–143. doi:10.1111/j.1095-8312.2012.01988.x

1. Ecological radiation with limited morphological diversification in salamanders.
Blankers, T., Adams, D.C. & Wiens, J.J. 2012. *Journal of Evolutionary Biology* 25: 634–646. doi:10.1111/j.1420-9101.2012.02458.x

TEACHING

University Teaching Qualification (BKO) awarded 8 February 2024

2025 – present	**Track coordinator (backstop)**, MSc Biological Sciences, General Biology track
2022	**Course design**: Current Topics in Biology, MSc Biological Sciences, General Biology track

Courses

2025 – present	Evolution of Plasticity and Learning · lecturer
2024 – present	Analysis & Modeling Lab, MSc Earth Sciences · module design, lecturer
2024 – present	Psychobiologie Literature Review · advisor
2023 – present	Van Neuron tot Gedrag · lecturer
2022 – present	Current Topics in Biology, MSc Biological Sciences · coordinator, lecturer
2022 – 2024	Keystone Project II: The Resilient Environment, BSc Biology · scientific advisor
2020 – present	Evolutiebiologie, BSc Biology · lecturer
2019 – present	Neurobiology and Behavior, BSc Biology · lecturer
2019 – present	Ecogenomics, BSc Biology · lecturer
2019 – 2020	Evolution and Behavior, BSc Biology · lecturer

Earlier teaching

2013 – 2015	Teaching assistant, Humboldt-Universität zu Berlin: MSc Bioacoustics (2014, 2015); BSc Aggressive Behavior (2013–2015)
2010 – 2012	Teaching assistant, University of Amsterdam: BSc Evolutionary Genetics (2012), Genetics and Evolution (2010); development of evolutionary genetics practicals for undergraduates (2012)
2004 – 2016	Homework class teacher, Het 4e Gymnasium Amsterdam (2015–2016); homework tutor, StudentsPlus (2008–2009); course in life-sciences education, UvA (2007, 12 EC); sailing and windsurfing instructor, WSC Sloterplas (2004–2012)

SUPERVISION

PhD students

2025 – present	Pietro Gadaleta · University of Amsterdam
2024 – present	Jiajia Zhao · University of Amsterdam
2018 – 2023	Elise Fruitet · University of Amsterdam, defended 27 September 2023
2016 – 2022	W. Hayden Waller · Cornell University, defended 16 September 2022

BSc, MSc and applied-university students

2025/26	Moya Lieffering (MSc, Leiden), Jesper Zonneveld (HBO, Hogeschool Leiden), Stan van Dijk (MSc), Octavio Cruces Guarderrama (MSc), Jeppe Vernooij (MSc)
2024/25	Charissa Claessens (MSc), Madelon Grim (MSc), Alex McKenna (MSc, Univ. Lille), Finn Verhoef (MSc), Daniel Nijman, Michelle Steenbergen, Monica Mooijer, Miel Houtenbos, Mattis Steenweg, Borai Lodewijk, Sophia Mulder (BSc), Maarten Ooijevaar (Leiden Univ. of Applied Sciences)
2023/24	Ellie Willson (MSc), Emma Diepeveen (MSc), Ihsen Lecat (MSc, Univ. Lille)
2022/23	Regina Marquez (MSc), Eléa Cavigneaux (MSc), Erdem Eren (BSc, Türkiye)
2021/22	Arie Ebbenhorst (Aeres Univ. of Applied Sciences), Jerry Schouten (BSc), Canberk Köroğlu (BSc, Gebze Technical University), Lotta Vaskimo (MSc, Leiden)
2020/21	Eline de Vries (MSc), Wout van der Heide (MSc, UvA & Cornell), Ekaterine Kikodze (BSc, Radboud University), Luka Rieger (MSc)

2018–2020	Thomas Rietbergen, Max Boot, Robin Moene (BSc), Arlet Culhaci (Leiden Univ. of Applied Sciences)
2013–2014	Carolin Geelhaar (MSc), Anna Lübke (BSc), Rafael Block (BSc) · Humboldt-Universität zu Berlin

University of Amsterdam unless stated otherwise.

INVITED AND CONTRIBUTED TALKS

Talks marked TALK ; other entries are conference contributions.

TALK Repeatability of evolution is additively affected by environmental and demographic variation. ESEB 2025

Eco-evolutionary feedbacks in *C. elegans*–microbe interactions. ESEB 2025

TALK De zoektocht naar buitenaards leven: de (on)begrensde mogelijkheden van leven op aarde. KNAW symposium 2025

KEYNOTE The role of behavior in shaping emergent ecological and evolutionary properties. Netherlands Society for Behavioural Biology 2024

TALK Eco-evolutionary dynamics in *C. elegans*–microbe interactions. WUR Laboratory of Genetics seminar 2024; EvoWorm 2024

TALK Boundaries on evolutionary predictions from selection and drift. NWO Life 2024

TALK Limits to evolutionary predictability through uncertainty in selection and drift. NLSEB 2024

TALK The predictability of evolution. Lorentz Workshop "Evolution of Complex Contemporary Life" 2024; Origins Center Conference 2023

TALK Universal patterns in speciation dynamics. NLSEB 2023

TALK Synthesizing causes and consequences of variation across sexual communication systems. NWO Life 2023

TALK Variation and function in sexual signal components. Entomology Day 2022

Experimental evolution of a pheromone signal. ESEB 2022

TALK Predictability of experimental evolution in *C. elegans*. NLSEB 2022

TALK Experimental host–microbiome evolution in *C. elegans*. NWO Life 2022

TALK Genetic, phenotypic, and biogeographic perspectives on population divergence in a Hawaiian cricket. ESEB satellite symposium on reproductive isolation 2021

TALK The selection response of a multicomponent sexual signal. Evolution (SSE) 2021

TALK Plasticity, variation, and constraint in sex pheromone signals. NLSEB 2021

TALK Convergent gut microbiota as a signature of non-adaptive speciation. NAEM 2021

TALK The predictability of genetic adaptation. Origins Center Conference 2021

TALK The evolution of sex pheromones. Netherlands Entomological Society, Ede 2019

Parallel genetics of repeated sexual signal divergence. ESEB, Turku 2019

Evolution of sex pheromone communication in response to artificial selection. NLSEB, Ede 2019

TALK A rapid radiation of flightless crickets in Hawaii. Netherlands Entomological Society, Ede 2018

Parallel genetic architectures for a behavioral speciation phenotype. ESEB, Montpellier 2018

TALK Convergence and constraints in behavioral isolation: a tale of two crickets. NBB symposium, Cornell University 2017

TALK The genomic architecture of repeated sexual signal divergence. Evolution, Portland 2017

The geographic context of speciation in crickets. Evolution, Portland 2017

TALK Sexual selection, population divergence, and the genomic mosaic of local evolutionary histories. EvoDay, Cornell Lab of Ornithology 2017

TALK Speciation islands? Behavioral and genomic divergence along a spatio-temporal cline. NBB symposium, Cornell University 2016

TALK Population genetics becomes population genomics: what the 'omics era means for evolutionary biology; and Divergence in acoustic and chemical mate signaling genes after speciation with gene flow in field crickets. BeGenDiv, Berlin 2015

Patterns of transcriptomic divergence and demographic history in field crickets. ESEB, Lausanne 2015

TALK The genetic architecture of calling song and song preference traits in *Gryllus rubens* and *G. texensis*. Orthopteran Bioacoustics Meeting, Zingst 2015

Premating reproductive isolation between two closely related cricket species. ESEB, Lisbon 2013

SERVICE AND LEADERSHIP

Boards and committees

Steering committee	Netherlands Society for Evolutionary Biology (NLSEB)	Mgmt. committee	COST European Network in Chemical Ecology (E-NICHE)
Steering committee	Dutch Origins Center	Member	UvA Scientific Advisory Committee
Steering committee	Amsterdam Center for Origins of Life (AMCOOL)	Member	European Society for Evolutionary Biology

Meetings and workshops organized

2026	Origins Center annual meeting
2024–2025	E-NICHE COST meetings and workshops: communication strategy (Amsterdam, Feb 2024), social media communication (Prague, Jul 2024), career development in chemical ecology (May 2025), annual meeting (Évora, Jun 2025)
2022–present	Host, weekly institute seminar ("Soup lunch"), IBED
2021	Quantitative Trait Mapping Workshop, Swammerdam Institute for Life Sciences, UvA
2020	Co-convener, Netherlands Annual Ecology Meeting
2015	Co-organizer, Bioacoustics meeting, Zingst, Germany

Peer review

About twice as many reviews per year as papers (co-)authored, for journals including *Evolution, Evolution Letters, Journal of Evolutionary Biology, Molecular Ecology, Proceedings of the Royal Society B, Trends in Ecology and Evolution, Ecology and Evolution, Behavioral Ecology and Sociobiology, Biological Journal of the Linnean Society, Zoological Journal of the Linnean Society, Ecological Entomology, Entomologia Experimentalis et Applicata, Systematic Entomology, Journal of Insect Science, Contributions to Zoology, Genes, PLOS ONE, F1000, PCI Evolutionary Biology* and Axios Review.

OUTREACH AND MEDIA

2025	KNAW symposium De zoektocht naar buitenaards leven · 360° VR lab video with Smart VR Lab (YouTube) · media interviews for EOS Wetenschap and nu.nl · Wetenschapsweekend Science Park · Meesterproef supervision, Maris College Den Haag
2023–2025	E-NICHE YouTube channel on chemical ecology
2024	ExpeditieNext Zutphen, Origins Center stand · Wetenschapsweekend Science Park
2023	Origins Center podcast "Over leven in het heelal", episode 5 · American Genetic Association blog · ExpeditieNext Middelburg · Onderzoeker voor de klas, Gemeentelijk Gymnasium Hilversum
2022	Boekenfest flash lecture "De geur van liefde", Oudemanhuispoort

2021	Radio interview, NPO Radio 1 *Fris*, on human chemical ecology · interview nu.nl, Kunnen we evolutie op lange termijn voorspellen? · primary school outreach, De Vuurvogel, Uithoorn
2014–2019	Betadag Regius College and school visit at Science Park Amsterdam (2019) · Amsterdam Green Campus workshop on energy transition (2019) · EEB & NBB Open Days, Cornell (2016, 2017) · Lange Nacht der Wissenschaften, Museum für Naturkunde Berlin (2014)

SKILLS AND LANGUAGES

Programming	R, Python, Bash	Languages	Dutch (native), English (C2), German (C1), Spanish (B2)

Updated November 2026 · Current version at thomasblankers.com